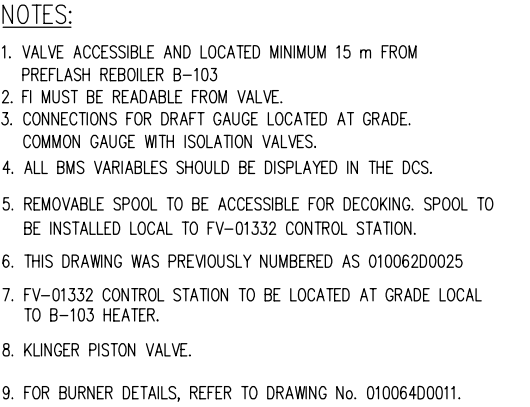




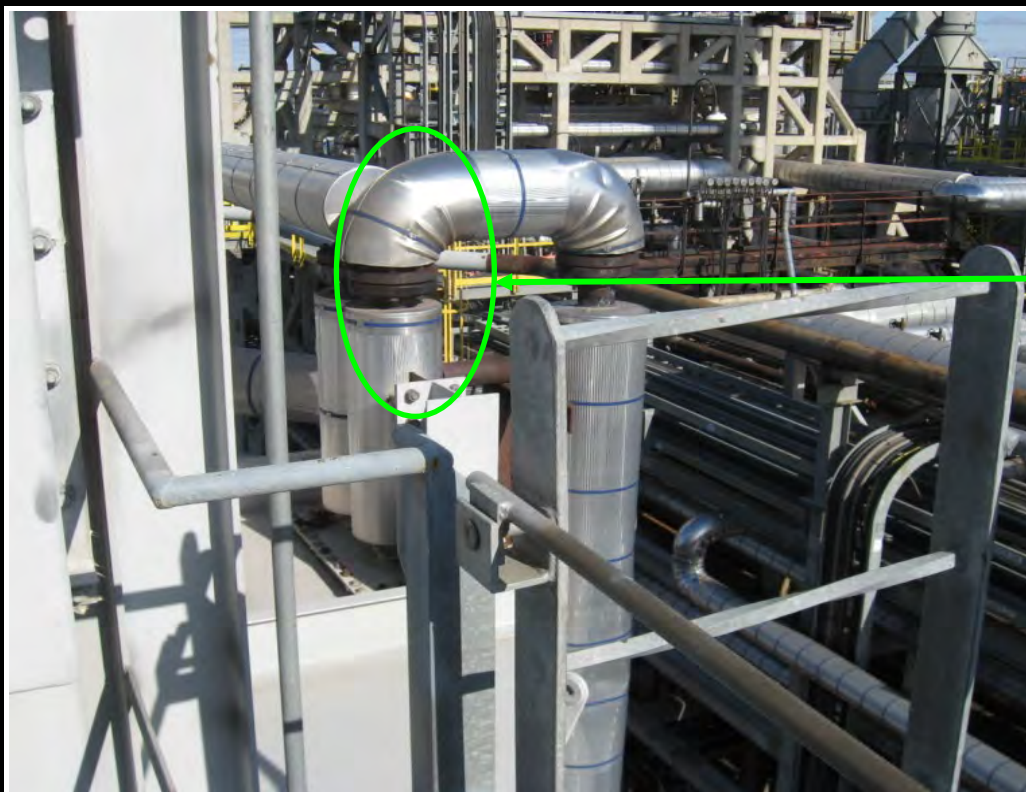
LEGENDE:

M01Xxxx N° ISOMÉTRIQUES D'INSPECTION

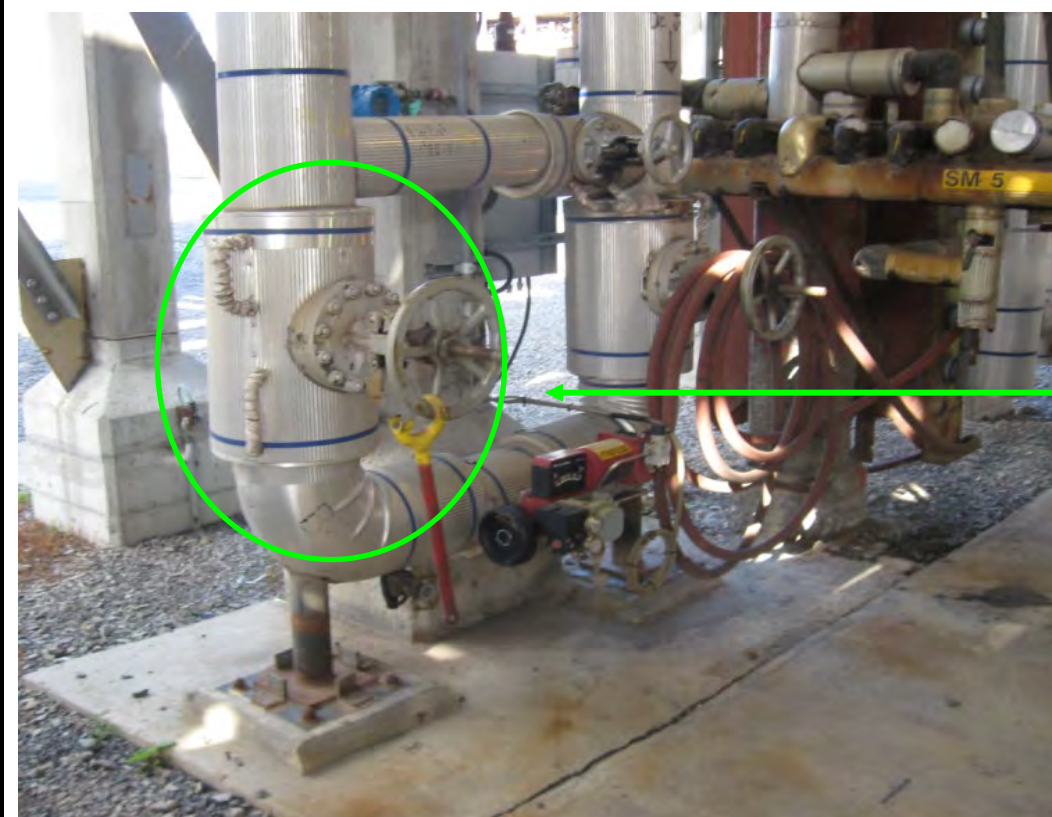
———— 5% Cr.

———— C.S.

BANTREL		TORONTO, ONTARIO	
		Job No. 956-010	
NO.	DESSIN DE REFERENCE		
		PETRO-CANADA MONTREAL, QUEBEC FABRICATION/MANUFACTURING	
CRUDE UNIT # 010 010/100 INTEGRATION B-103			
SCHEMA TUYAUTERIE / INSTRUMENTATION			
DESSEIN/	B.P.	DATE/	JAN 2003
CONCEPTEUR/		VERIFIE/	S.S.
		NO. DU DESSIN/	010062D0035
REV.	1=1		
ECHELLE/	N.T.S	CLASSE/	G
			010-062D0035
			4



Exit 4"-#300 on the 4th balcony of the furnace. Temporary spool to be installed by suncor.
8 ft/square surface area.



Entrance 4"-#300 at ground level. Temporary spool to be installed by suncor.
10 ft/square surface area.

CODE DE SERVICE: B-0103

Pre-Distillation Reboiler furnace

AVIS#

ORDRE#

* Furnance service Crude Pre- distillation

* All temporary spools and taking out all instruments on decoke piping to be done by suncor personnel.

* Entrance is at groud level. The receiver is on the top balcony of the furnace elavation 35 ft high. balcony 8ft x 5ft

RÉV:

00



Petro-Canada Products
Produits Petro-Canada
A division of Petro-Canada Inc.
Une division de Petro-Canada Inc.

SAP#

100097011

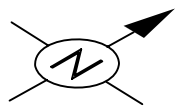
CODE DE SERVICE

M01B077

DESCRIPTION

BRUT PRÉDÉTENTE DU J0132/A/B VERS B0103

(O-1923-4"-UD1)

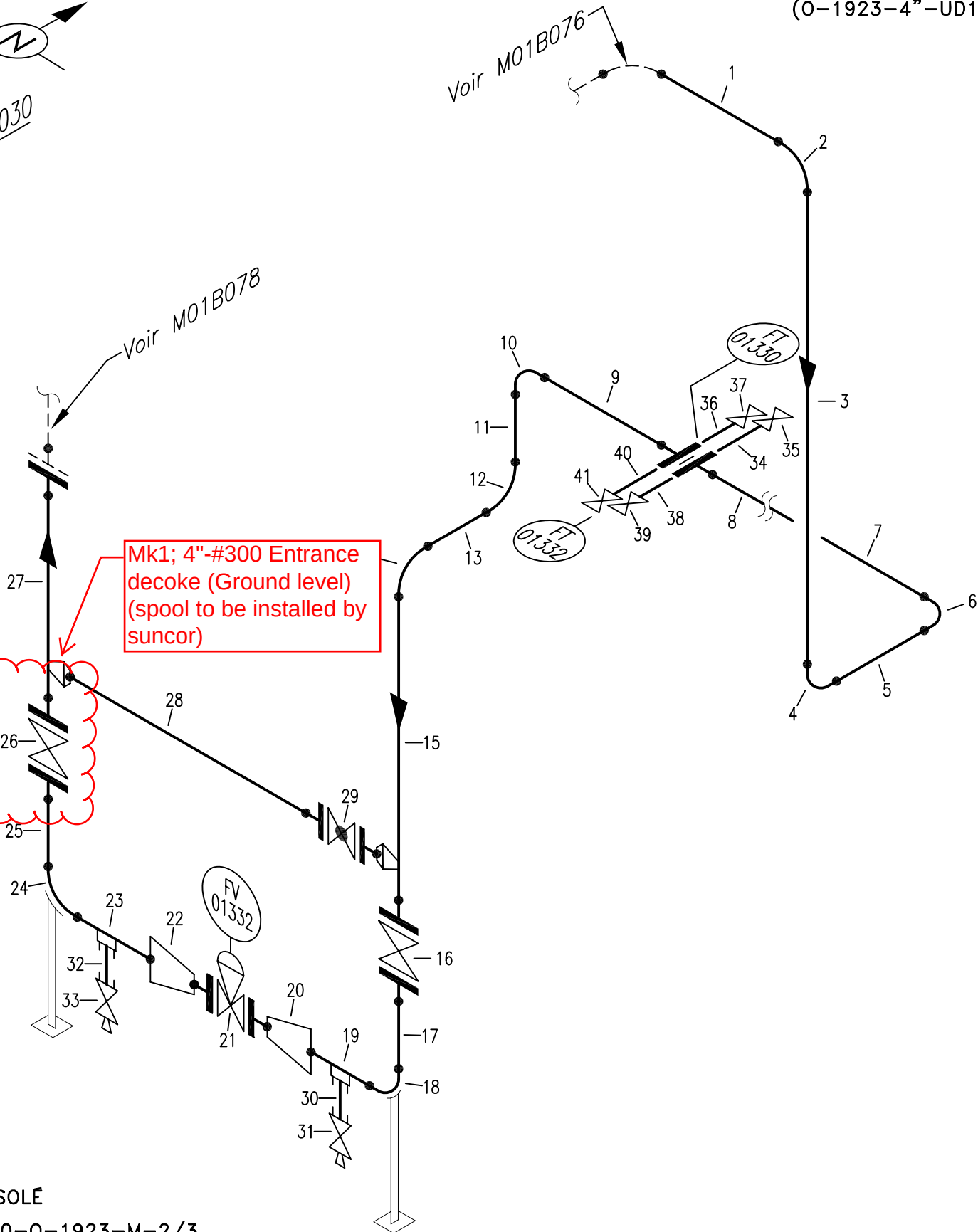
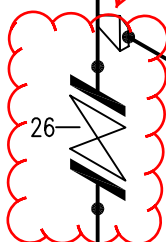


5030

Voir M01B076

Voir M01B078

Mk1; 4"-#300 Entrance
decoke (Ground level)
(spool to be installed by
suncor)



ISOLÉ

010-O-1923-M-2/3

DEC. 2005

D.M.

Rev.0



SAP#

302020345

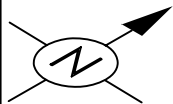
CODE DE SERVICE

MO1FO24

DESCRIPTION

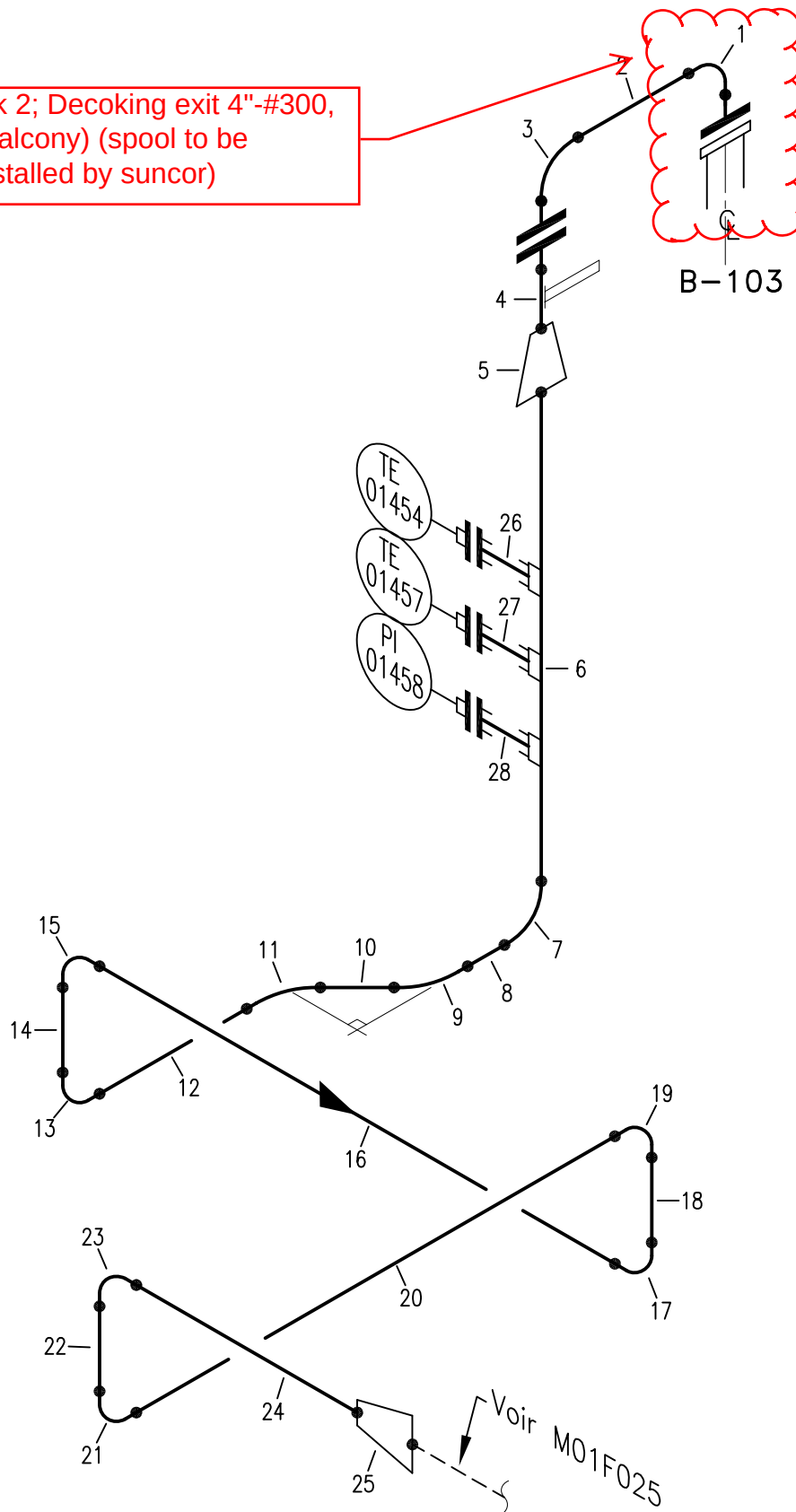
BRUT PRÉDETENTE DU B0103 VERS E0131

(0-1925-6"/4"-UD10)



1640

Mk 2; Decoking exit 4"-#300,
(Balcony) (spool to be
installed by suncor)



ISOLÉ

010-O-1925-M-1/2

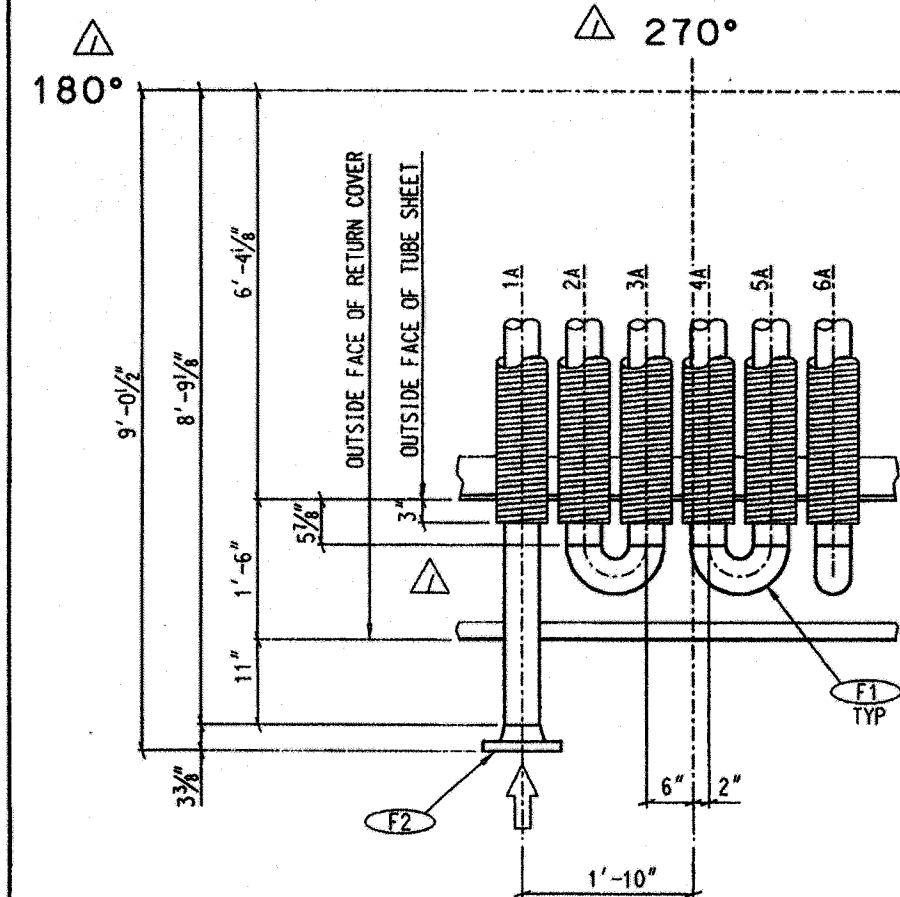
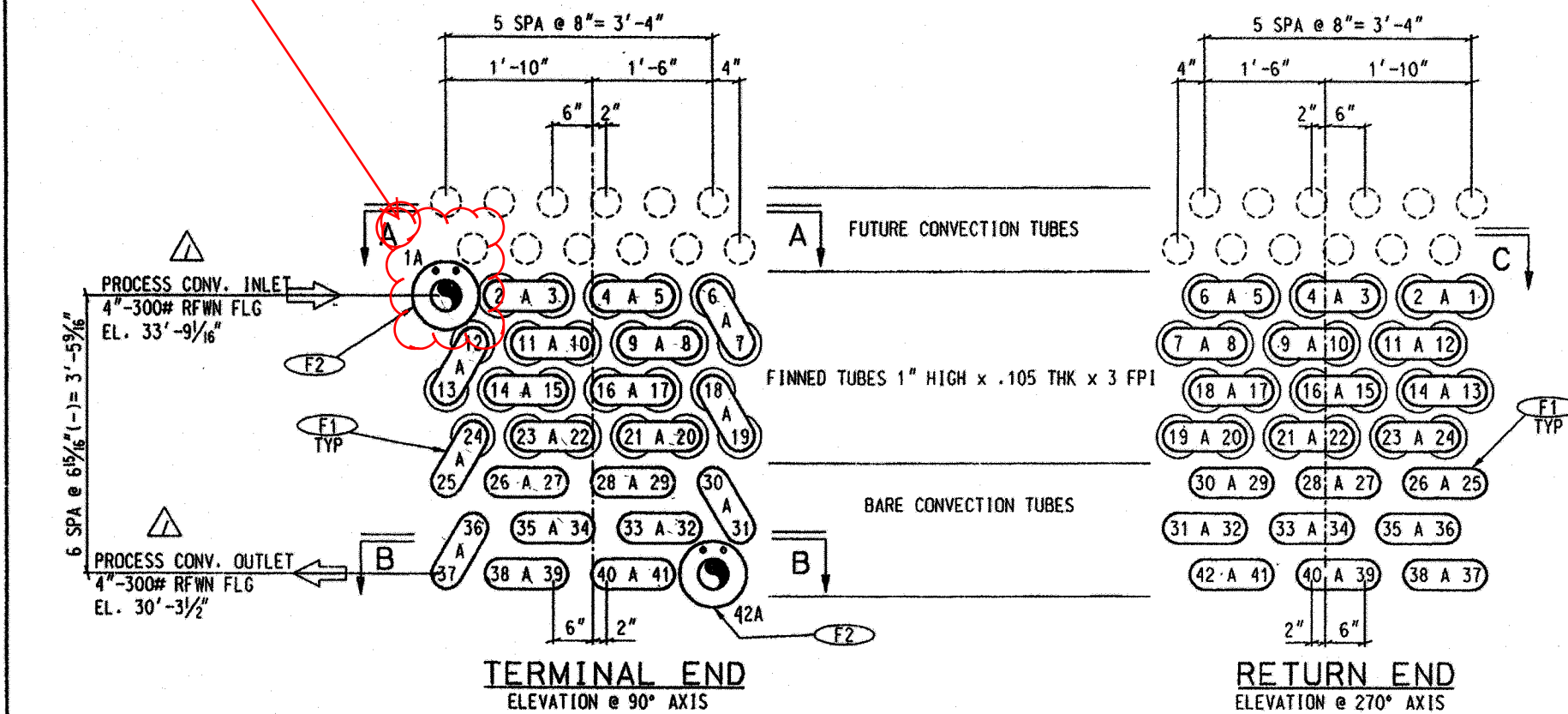
RED. 09/06

D.M.

Rev.0

Entrance
convection 4"-#300

16

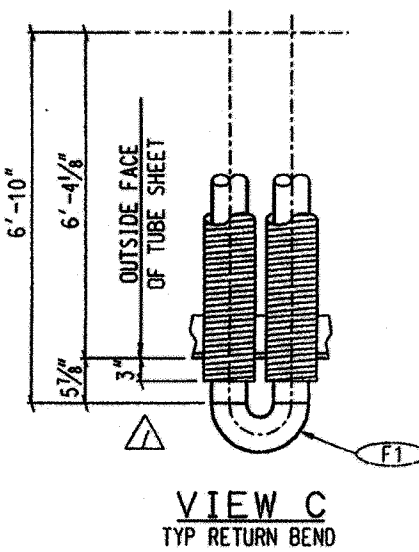


NDE REQUIREMENTS

RT: 100% of 100%
UT: NR
MT: NR
PT: NR
PMI: 100%
BHN: 220 BHN

PROPRIETARY INFORMATION

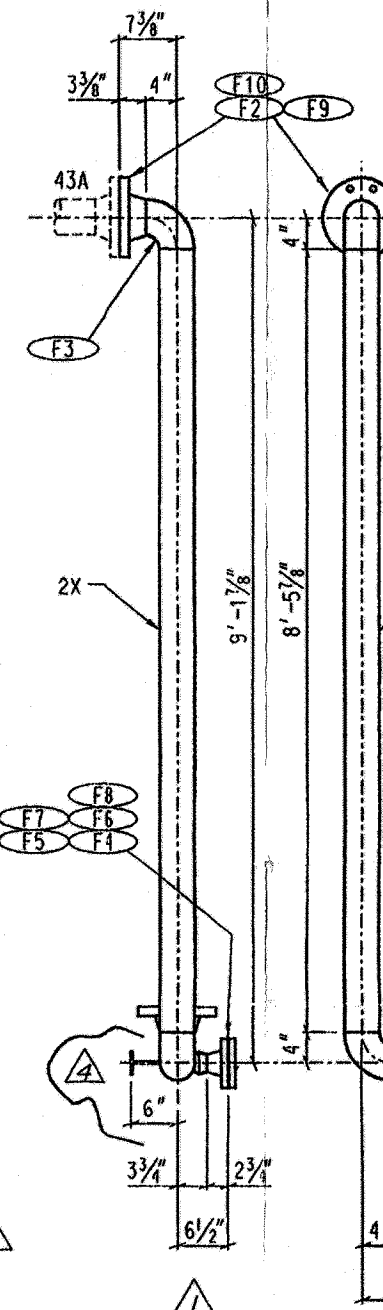
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Reviewed	DATE: 6/10/04
RE: 2042	

VIEW B - B
PLAN @ CONVECTION OUTLET

Bantrel	Job No. 956-010
SUPPLIER DOCUMENT STAMP	
1. WORK MAY PROCEED.	
2. REVISION AND RESUBMIT. WORK MAY PROCEED SUBJECT TO INCORPORATION OF CHANGES INDICATED.	
3. STOP WORK PER ATTACHED WRITTEN INSTRUCTIONS.	
4. REVIEW NOT REQUIRED. WORK MAY PROCEED.	



ONE CROSSOVER Mk. 4A-1
SHIP LOOSE TO FIELD

NOTES:

- ALL ALLOY TUBES, PIPE AND FITTINGS WILL BE 100% PMI TESTED BY TULSA HEATERS, INC.
- ALL COIL MATERIAL MUST BE PROTECTED FROM PAINT AND ZINC CONTAMINATES DURING STORAGE, FABRICATION AND INSTALLATION. TUBES SHALL NOT COME IN CONTACT WITH GALVANIZED METAL AT ANYTIME. NO MATERIAL IS TO BE LEFT UNPROTECTED.
- ALL WELD PROCEDURES ARE TO BE APPROVED PRIOR TO START OF WELDING.
- PREHEAT AND POSTWELD HEAT TREATMENT TO BE IN ACCORDANCE WITH APPROVED WELD PROCEDURES.
- RADIOGRAPHY TO BE PERFORMED AFTER COMPLETION OF POSTWELD HEAT TREATMENT.
- FLANGE FACE FINISH OF ALL FLANGED CONNECTIONS ON THE PROCESS (PREFLASH REBOILER) COIL SHALL BE 125 AARH MAX.

WORK THIS DWG WITH DWG 4B

PROCESS COIL DATA

OPER. CONDITION:
INLET 420.7° F. @ 152 PSIG
OUTLET 640° F. @ 40 PSIG
DUTY 17.58 MM BTU/HR.

DESIGN CONDITIONS:
PRESSURE: 450 PSIG
FLUID TEMPERATURE: 690° F.
CONV TUBE WALL TEMPERATURE: 749° F.
RAD TUBE WALL TEMPERATURE: 900° F.
CORROSION ALLOWANCE: 0.125"
SHOP HYDROTEST @ 1125 PSIG

CUSTOMER:
BANTREL CO./PETRO-CANADA
PROJECT : MONTREAL CRUDE RECONFIGURATION
LOCATION : MONTREAL, QUEBEC, CANADA
UNIT : CRUDE UNIT 10
SERVICE : PREFLASH REBOILER
EQUIP. No. : B-103
P.O. No. : 956-010-010-F-01

Rev.	Date	By	Revision Description
10/9	BP	REV'D.	PER CUSTOMER COMMENTS
12/9	JP	REV'D.	PER CUSTOMER COMMENTS
2/3	JP	REV'D.	OPER & DESIGN CONDITIONS
2/26	JP	REV'D.	PER CUSTOMER COMMENTS
8/2	JP	REV'D.	PER CUSTOMER COMMENTS

TUBE & PIPE BILL of MATERIAL

REV	MARK	QTY	FIN	DESCRIPTION SIZE x WALL	LENGTH FT IN	END PREP	MAT'L	LOCATE REMARK
1A	ONE	YES	4" SCH 40 PIPE	15 7/8	BEV	A335	PROCESS	
2A-24A	23	YES	4" SCH 40 PIPE	13 8	BEV	A335	CONVECTION	
25A-41A	17	NO	4" SCH 40 PIPE	13 8	BEV	A335	CONVECTION	
42A	1	NO	4" SCH 40 PIPE	15 7/8	BEV	A335	CONVECTION	
43A, 88A	2	NO	4" SCH 40 PIPE	26 4 1/2	BEV	A335	RADIANT	
44A-87A	44	NO	4" SCH 40 PIPE	21 9	BEV	A335	RADIANT	
1X	ONE	NO	4" SCH 40 PIPE	6 11	BEV	A335	CROSS	
2X	ONE	NO	4" SCH 40 PIPE	8 5 1/2	BEV	A335	CROSS	

FITTINGS BILL of MATERIAL

REV	MARK	QTY	DESCRIPTION	MATERIAL	LOCATE REMARK
F1	41	4" SCH 40 180° S.R. RETURN BENDS, 10 TO MATCH SCH 40 (1/4-12/8)	A234-WP5	CONVECTION	
F2	4	4"-300# RFWN FLG (SCH 40 BORE)	A182-F5	CONVECTION	
F3	3	4" SCH 40 90° S.R. ELBOW, 10 TO MATCH SCH 40 (1/4-12/8)	A234-WP5		
F4	ONE	2"-300# RFWN FLG (SCH 40 BORE)	A182-F5		
F5	ONE	2"-300# BLIND FLG	A182-F5		
F6	ONE	2"-300# FLEXITALLIC GASKET (CG-3G)	304SS		
F7	8	5/8" x 3 1/2" LG STUD w/(2) HEX NUTS EA	A193-B7/A194-2H		
F8	ONE	4" RUN x 2" OUTLET WELDOLET (STD WT)	A234-WP5		
F9	2	4"-300# FLEXITALLIC GASKET (CG-3L)	304SS		
F10	16	3/4" x 4 1/2" LG STUD w/(2) HEX NUTS EA	A193-B7/A194-2H		
F11	2	4"-300# RFWN FLG (SCH 40 BORE)	A182-F9	RADIANT	
F12	2	PL 3/8" x 6" x 0'-6 1/2" LG	310SS		
F13	23	4" SCH 40 180° S.R. RETURN BENDS, 10 TO MATCH SCH 40 (1/4-12/8)	A234-WP9	RADIANT	
F14	22	4" SCH 40 180° L.R. RETURN BENDS, 10 TO MATCH SCH 40 (1/4-12/8)	A234-WP9	RADIANT	
F15	23	1" x 1'-4" LG (T.O.E.)	310SS		
F16	23	STD FLAT WASHER FOR 1" BOLT	CS		
F17	23	HVY HEX NUT FOR 1" BOLT (GALV.)	A307		
F18	23	6" RUN x 3/4" OUTLET SOCKLET (STD WT)	A234-WP9		
F19	1	PL 1/4" x 3" x 0'-3"	304SS		
F20	1	PL 1/4" x 3" x 0'-3 1/2"	304SS		
F21	1	PL 3/8" x 6" x 0'-8"	304SS		
F22	1	PL 3/8" x 6" x 0'-7 1/2"	304SS		
F23	2	PL 3/8" x 3 1/2" x 0'-8 1/2"	304SS		
* SHIP LOOSE TO FIELD					

TUBE FINNING BILL of MATERIAL

NOTES:
1. All fins helically wound. "N" Fins/
root. "H" High, and "T" Thick.
2. Fin Types are as follows:
SOLID = Solid Fins &
SER = Serrated Fins.
3. All fins to be continuous weld.
High Frequency.

REV	MARK	QTY	TUBE MAT'L	TUBE O.D.	TUBE WALL	"L" FT/IN	FIN TYPE	"A" FT/IN	END PREP	"B" FT/IN	"C" FT/IN	END PREP	"N" FT/IN	"H" FT/IN	"T" FT/IN	FIN MAT'L
1	1A	1	A335 P5	4 1/2	SCH 40	15 7/8	SOLID	2 2	BEV	13 2/4	0 2/4	BEV	36	105	1	409SS
1	2A-24A	23	A335 P5	4 1/2	SCH 40	13 8	SOLID	0 2/4	BEV	13 2/4	0 2/4	BEV	36	105	1	409SS
									BEV			BEV				
									BEV			BEV				
									BEV			BEV				
									BEV			BEV				
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SHOULDER
REQUIREMENTS
CONTROLLED
DOCUMENT CONTROL

17 SEP. 2004

015837

PROJECT
CONTR.P.O./REF
Q10.FOL-4A-2004

TULSA
HEATERS
INC.

COIL DETAILS

Drawn By: JP Date: 7-25-03 Job No.: 03420
Checked By: PW Date: 7-29-03 Drawing No.: 03420-4A 5